

DLM RESEARCH REPORT

Neural Signature Classification using Universal EEGNet Framework

Executive Summary

This technical report documents the development and validation of a **Deep Learning Model (DLM)** designed to identify neural recognition patterns across a diverse multi-subject population. By utilizing a 16-channel sensory array, the DLM achieves high sensitivity in detecting universal brainwave deflections, specifically the P_{300} event, without requiring individual calibration.

84.38%

TOTAL ACCURACY

86%

SENSITIVITY

10

SUBJECTS

Technical Methodology

The system utilizes the **EEGNetv4** architecture. Unlike traditional SVM classifiers, this DLM maintains the spatial integrity of the sensor grid, allowing for 2D convolutional filters to extract features from both time and electrode location simultaneously.

Scientific Analysis

1. Spatial Mapping

Analysis of the spatial convolutional layers confirmed that the model prioritizes **Channels 11 and 12**. This aligns with neurophysiological expectations for visual-related recognition tasks, as these sensors correspond to the parietal/occipital cortices responsible for visual processing.

2. Temporal Classification

The DLM effectively isolates positive voltage peaks occurring at the *~300ms* latency. The model has been tuned with weighted loss to ensure high recall, prioritizing the identification of "Target" recognition events over background stochastic noise.

Performance Metrics

Metric	Non-Target (Baseline)	Target (Recognition)
Precision	0.97	0.52
Recall (Sensitivity)	0.84	0.86
F1-Score	0.90	0.65

Future Scope

The universal nature of this DLM allows for immediate deployment in communication assistive devices, mental workload monitoring, and industrial safety systems where hands-free intent recognition is critical.